

25 YEARS IMPLEMENTATION 5 YEARS CAPITULIZATION 150 HECTARS

SOIL EMISSIONS

In the case it's LINEAR the amount of hectares per year is $\rightarrow 150/25 = 6$ hectares/year. So in 20 years $\rightarrow 20 \cdot 6 = 120$

After 20 we gain 6 more hectares and "lose" 6 hectares that are past 20 years of calculation. So hectares remain constant

After 25 (i.e. end of implementation) we only "lose" 6 per year, so by 30 years $150 - 5 \cdot 6 = 90$ TOTAL

$$\frac{20 \cdot 120}{2} = 1200$$

$$5 \cdot 120 = 600$$

$$(5 \cdot 90) + \frac{5 \cdot 30}{2} = 525$$

TOTAL
2325

FORMULA FOR SOIL EMISSIONS ON EXCEL

AREA - DEFORESTED $\cdot \Delta C_{\text{MINERAL}} / (\text{h} \cdot \text{YEAR}) \cdot \text{TIME COMPONENT}$

So to calculate yearly we have to just multiply $\Delta C_{\text{MINERAL}} / (\text{h} \cdot \text{YEAR})$ by the total number of hectares across all years.

So what we get as a result from our calculations is:

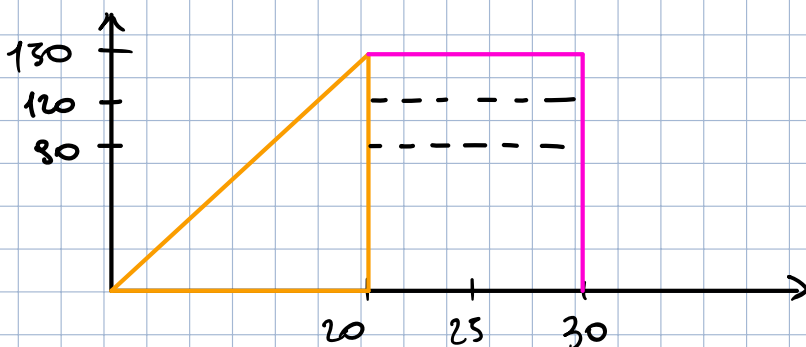
$$\text{AREA} \cdot \text{TIME COMPONENT} = \text{SOIL EMISSIONS} / \Delta C_{\text{MINERAL}} / (\text{h} \cdot \text{YEAR})$$

But with the EXCEL $\rightarrow = 2625$ WHY?

THE EXCEL FORMULA:

$$\begin{aligned} & \text{TOT_AREA} \cdot \left[\min(20, \text{time_impl}) \cdot \text{rate of change soil} \right] + \text{ } \\ & \text{TOT_AREA} \cdot \left[\text{time_tot} - \min(20, \text{time_impl}) \right] - \text{ } \\ & \text{TOT_AREA} \cdot \left[(\text{time_tot} - 20)^2 \cdot \frac{1}{2 \cdot \text{time_impl}} \right] \end{aligned}$$

MORE COMPLICATED TO DRAW, BUT...



$$150 \cdot [20 \cdot 0,3] = 1500$$

$$150 \cdot [30 - 20] = 1500$$

WHAT DOES IT REPRESENT?

$$150 \cdot \left[100 \cdot \frac{1}{2 \cdot 25} \right] = 300 \quad \sim 300 = \begin{cases} 5 \cdot 60 \\ 10 \cdot 30 \end{cases}$$

